Personalized Grocery and Diet Planning Application

Course: CP317

Group Number: 3

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1. Product Backlog

Present in an excel sheet attached to this submission.

2. Requirements Refinement

The Personalized Grocery and Diet Planning Application will bring grocery management, meal planning, and nutrition tracking together in one place. Users will be able to plan meals around the ingredients they already have while considering their dietary preferences and goals. The application will also help users monitor their nutritional intake and review their progress over time. These features aim to make daily food decisions easier and reduce the effort involved in managing several separate tools.

The following functional and non-functional requirements build on the objectives and user stories outlined in Milestone 1, and the expanded product backlog for Milestone 2.

Functional Requirements

1. **User Profile and Goal Setup**: Users will be able to create and update a profile with their dietary goals, personal information, and food preferences, including favourite cuisines, preferred foods, and foods they dislike. They will also be able to enter an optional weekly grocery budget to guide meal and grocery suggestions. Users may skip optional onboarding questions and use the application with clearly labelled generic settings.

2. **Dietary Preferences and Restrictions**: Users will be able to enter and update their dietary preferences, allergies, and food restrictions. The application will consider these restrictions when suggesting meals and exclude ingredients identified as unsuitable. If ingredient information is incomplete, the application will notify users that it cannot fully check whether the meal meets their restrictions.

3. **Nutrition Target Calculation**: The application will calculate estimated daily calorie and nutrient targets using the user's age, weight, height, activity level, and dietary goals. These targets will include protein, carbohydrates, fat, and fibre. Users will be able to compare their logged intake with their daily targets. When the information needed for personalized calculations is missing, the application will provide clearly labelled generic settings.

4. **Grocery Inventory Management**: Users will be able to add, edit, and remove grocery items from their inventory. Each entry will include the item's name, available quantity, and nutritional information where provided. The application will use this inventory to suggest meals that make use of ingredients users already own.

5. **Personalized Meal Suggestions**: The application will suggest meals based on the user's goals, preferences, dietary restrictions, and available groceries. When ingredients are limited, it will prioritize meals that use what the user already has and require few

additional purchases. Users will be able to review suggestions and choose meals for their plan. Each suggestion will show the required ingredients, identify missing items, and allow users to add them to their grocery list.

6. **Grocery List Generation**: The application will generate a grocery list from the user's selected meals, taking their existing inventory into account. Meal selections will reflect the food preferences recorded in their profile. The list will show the additional ingredients and quantities needed. If a grocery budget is provided and estimated ingredient prices are available, the application will display an estimated total and flag when it exceeds the budget. Users will be able to adjust the list to suit their needs.

7. **Meal Logging and Nutrition Tracking**: Users will be able to log their meals and the quantities consumed. The application will calculate the calories and nutrients in each entry using the available nutritional information. Daily summaries will show the user's total intake and how much remains or exceeds their targets.

8. **Progress Tracking, Streaks, and Historical Reports**: The application will store previous meal logs so users can review their nutritional intake and progress over time. It will also track meal-logging streaks, with a streak representing consecutive days on which a user records at least one meal. This feature will encourage consistent logging. Where users record grocery spending, the application will provide summaries to help them understand their spending habits

Non-Functional Requirements

1. **Privacy and Security**: The application will protect personal information, including profile details, dietary restrictions, and meal records. Authentication and access controls will limit access to each user's private data. Only information needed to support the application's features will be collected, and optional personal details will remain the user's choice.

2. **Usability and Accessibility**: The application will use a straightforward layout with readable text and clearly organized menus, making it easy for users to find features and complete everyday tasks. Nutrition summaries and progress displays will include written labels and numerical values so users can understand them without depending on colour. Forms and instructions will use plain language to support users with varying levels of experience.

3. **Performance and Responsiveness**: Common actions, such as updating groceries, saving a meal, and opening a daily summary, will respond promptly under normal prototype conditions. When a task takes longer, such as generating meal suggestions, the application will display a loading indicator to keep users informed.

4. **Accuracy and Reliability**: The application will use consistent units and defined formulas when calculating nutritional values. It will check numerical entries and flag missing or invalid information to reduce errors. Saved records will remain available across sessions.

Nutrition targets and meal recommendations will be clearly presented as estimates intended to support general dietary planning.

3. Ethical Considerations

1. **Data Privacy:** The application collects personal information during onboarding, including dietary goals, preferences, and potentially sensitive health-related information. To protect user privacy, onboarding (PROF-01) will make personalization optional, allowing users to skip questions without losing access to core functionality. Users who choose not to provide personal information will still receive recommendations based on reasonable default settings, although these may be less personalized. Additionally, users can modify or remove their dietary preferences and restrictions through PROF-03, providing greater control over their stored information.

2. **Security:** The application will incorporate security measures to protect personal information collected through onboarding and stored in user profiles (PROF-01, PROF-03). Sensitive user information will be encrypted when stored rather than maintained in plaintext, reducing the risk of exposure in the event of unauthorized database access. Additionally, appropriate authentication and access controls will be implemented to ensure that users can only access their own personal information.

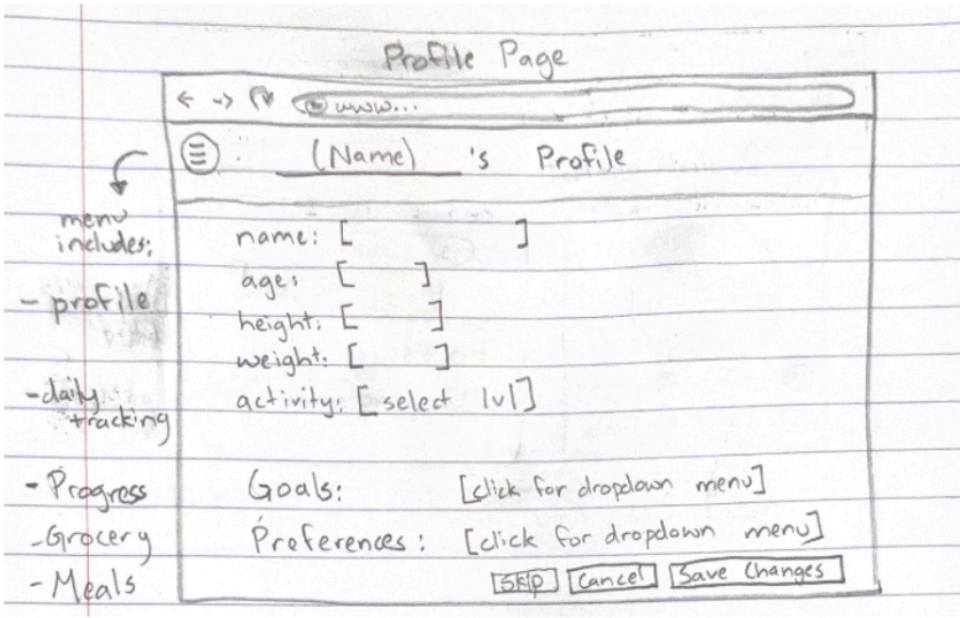
3. **Fairness and Inclusivity:** The application's personalization features will be designed to accommodate users with diverse dietary needs and backgrounds. Through PROF-03, users can specify dietary preferences, restrictions, and allergies, with flexible options beyond predefined categories where feasible. These preferences will inform grocery recommendations (GROC-01,02) and meal suggestions (MEAL-01,02), ensuring recommendations respect individual dietary requirements. Additionally, optional onboarding (PROF-01) ensures users who choose not to share personal information can still access core features.

4. **Accessibility and Usability:** The application's GUI will incorporate high-contrast text, readable font sizes, clearly labelled input fields, and intuitive navigation, particularly during onboarding (PROF-01) and meal and grocery planning (MEAL-01,02 & GROC-01,02). Nutritional information will be displayed using numerical values and text labels alongside visual indicators rather than relying solely on colour. These measures will help ensure that users with varying visual abilities and levels of technical experience can effectively interact with the application.

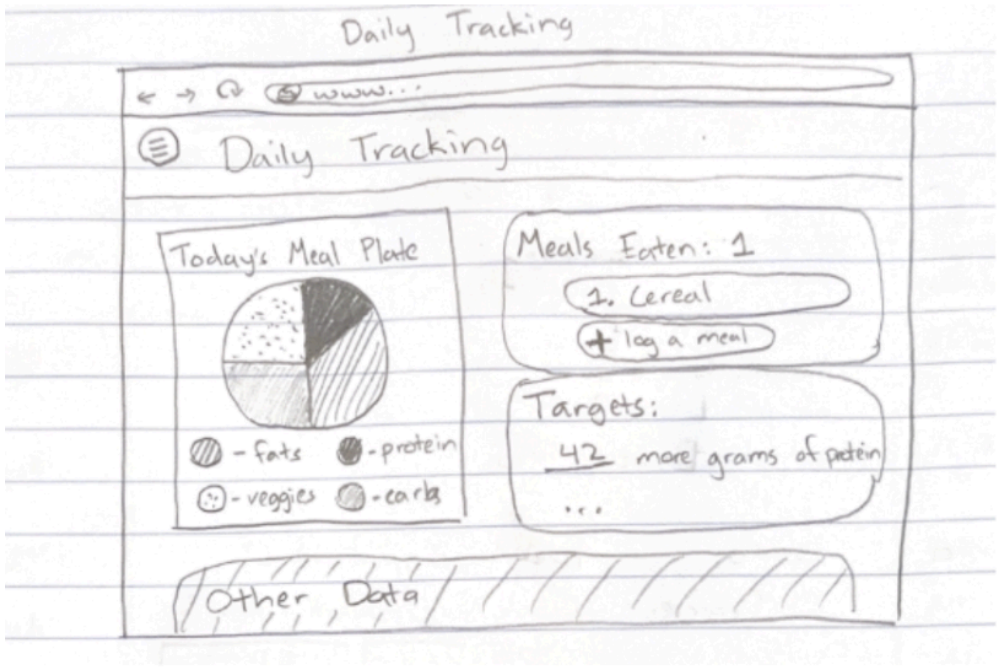
5. **Accuracy and User Safety:** The nutritional target calculations (PROF-02) will use established formulas and undergo testing to verify their accuracy. Users will also be able to update their personal information and goals, allowing calculated targets to reflect

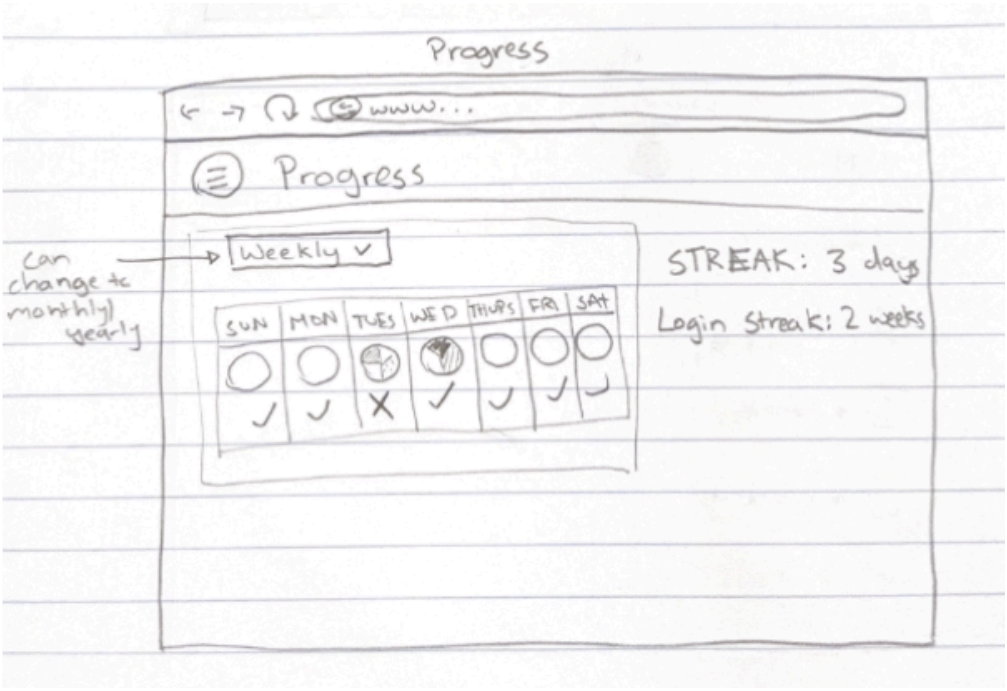
changes in their circumstances. Additionally, grocery and meal recommendations (MEAL-01,02 & GROC-01,02) will be designed to respect specified dietary restrictions and allergies. Automatically generated nutritional targets will be presented as estimates, and users will be informed that the application provides general nutritional guidance rather than professional medical advice.

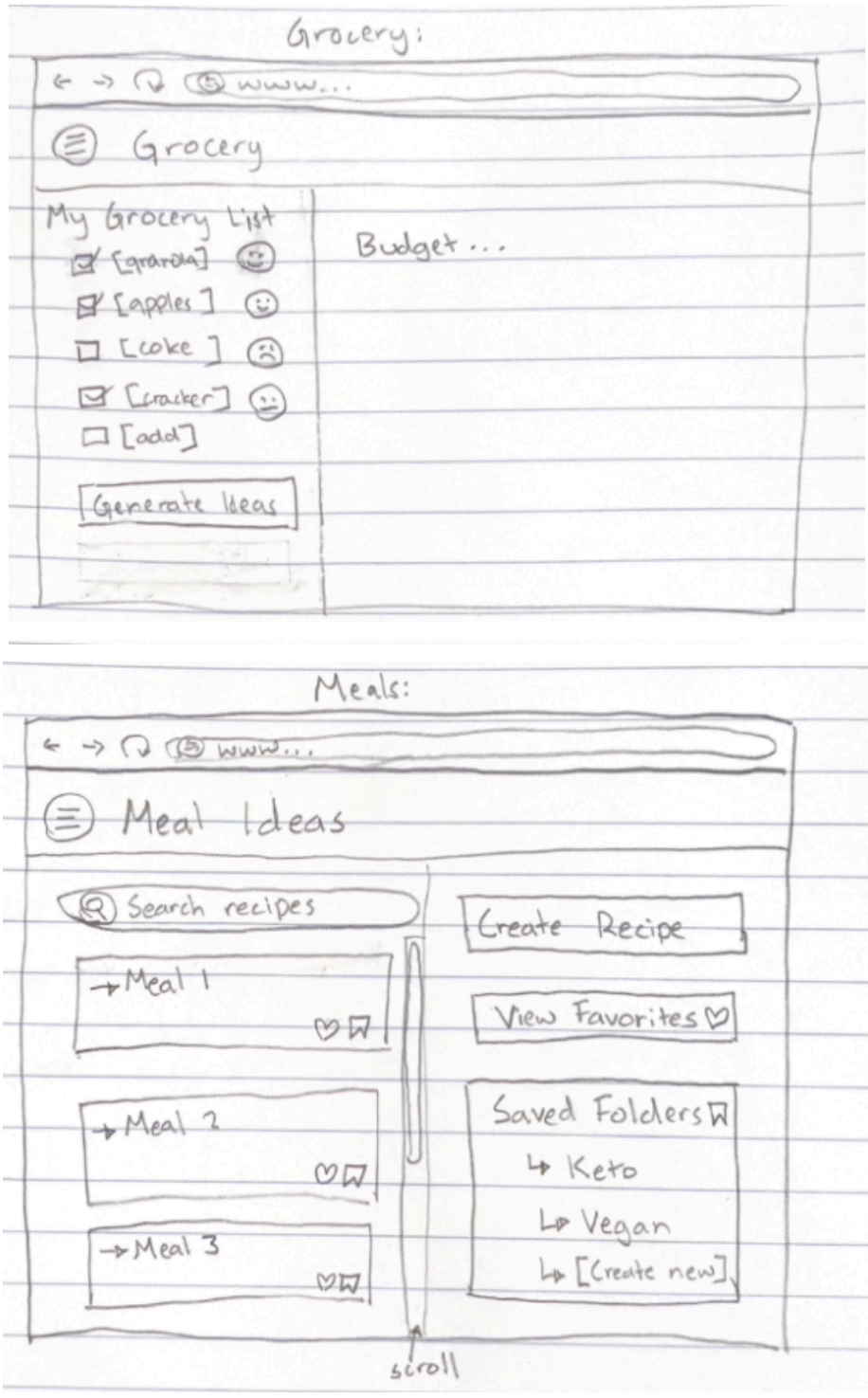
4. Initial Design Sketches






Daily Tracking
www...
Daily Tracking
Today's Meal Plate
- fats
- protein
- veggies
- carbs
Meals Eaten: 1
1. Cereal
+ log a meal
Targets:
42 more grams of protein
...
Other Data


Progress
www...
Progress
can change to monthly yearly
Weekly
SUN
MON
TUES
WED
THURS
FRI
SAT
STREAK: 3 days
Login Streak: 2 weeks



5. Team Blog

Present in an excel sheet attached to this submission.